

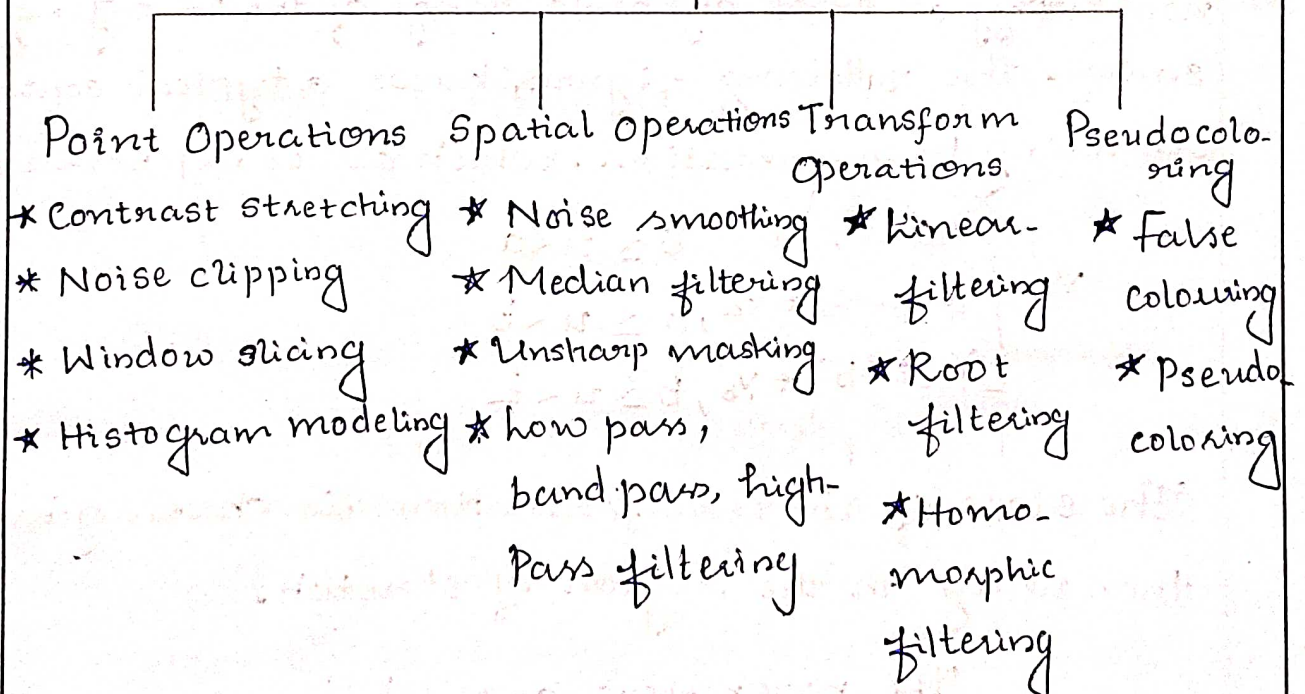
MODULE - 3

IMAGE ENHANCEMENT IN SPATIAL & FREQUENCY

DOMAIN

Image enhancement refers to accentuation, or sharpening, of image features such as edges, boundaries, or contrast to make a graphic display more useful for display and analysis. Some of the common image enhancement techniques.

Image Enhancement



POINT OPERATIONS

Point operations are zero memory operations where a given gray level $u \in [0, L]$ is mapped into a gray level $v \in [0, L]$ according to a transformation.

$$v = f(u)$$

CONTRAST STRETCHING

Low-contrast images occur often due to poor or uniform lighting conditions or due to nonlinearity or small dynamic range of the imaging sensor. The following figure shows a typical contrast stretching transformation, which can be expressed as

$$v = \begin{cases} \alpha u, & 0 \leq u < a \\ \beta(u-a) + v_a, & a \leq u < b \\ \gamma(u-b) + v_b, & b \leq u < L \end{cases}$$

The slope of the transformation is chosen greater than unity in the region of stretch.

The parameters a and b can be obtained by examining the histogram of the image.

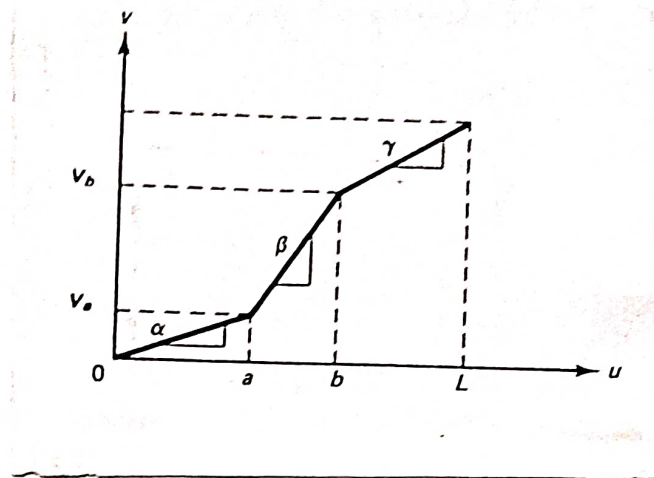
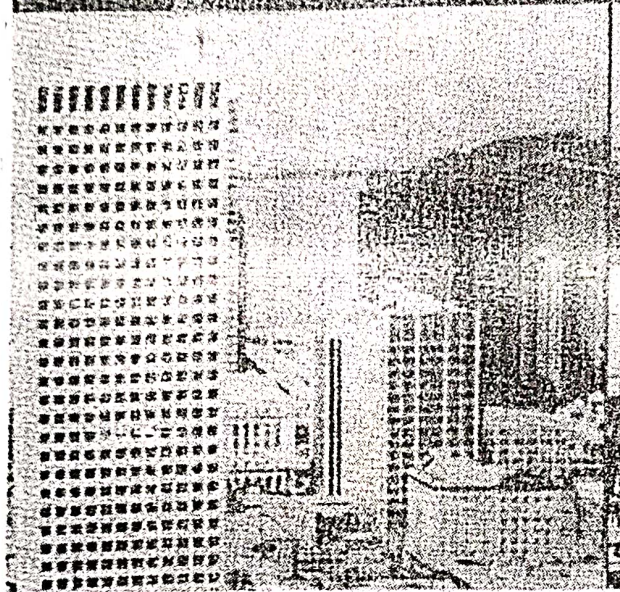
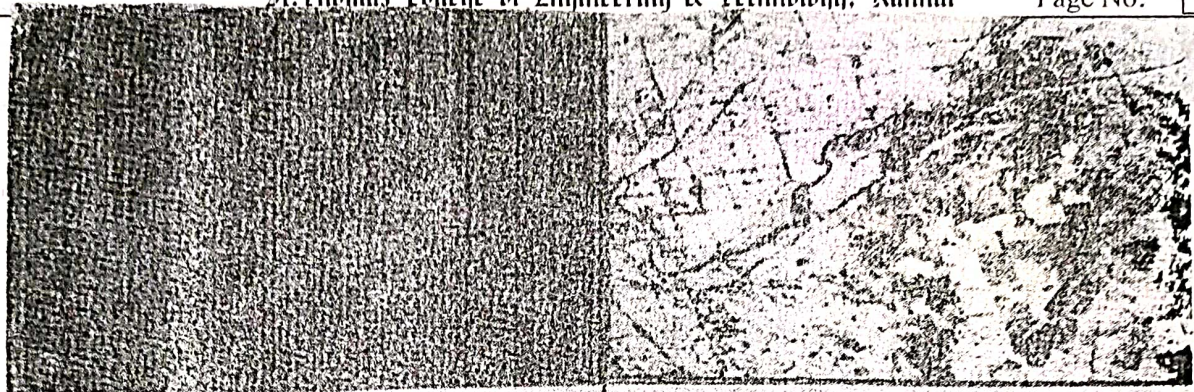


Figure 1: Contrast stretching transformation. For dark region stretch $\alpha > 1$, $a \approx L/3$; mid region stretch, $\beta > 1$, $b \approx \frac{2}{3}L$; bright region stretch $\gamma > 1$.

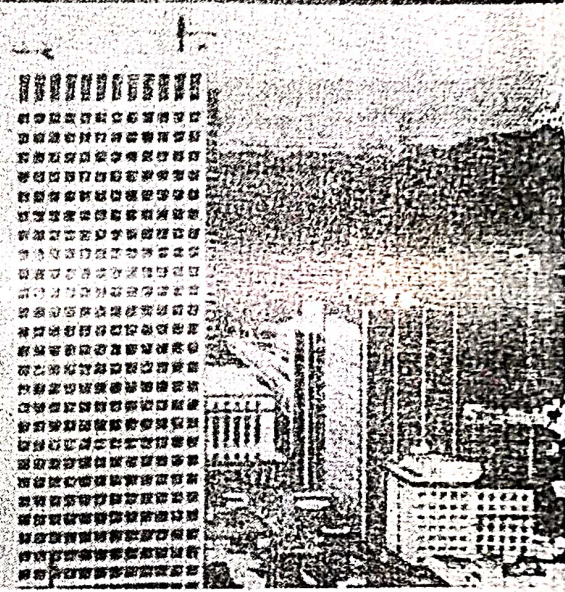
CLIPPING AND THRESHOLDING

A special case of contrast stretching where $\alpha = \gamma = 0$ (Figure 3) is called clipping. This is useful for noise reduction when the input signal is known to lie in the range $[a, b]$.

Thresholding is a special case of clipping where $a = b \triangleq t$ and the output becomes binary (Figure 4). Figure 5 shows examples of clipping and thresholding.



Originals



Enhanced



Originals

Enhanced



Originals

Enhanced

Figure 2: Contrast Stretching

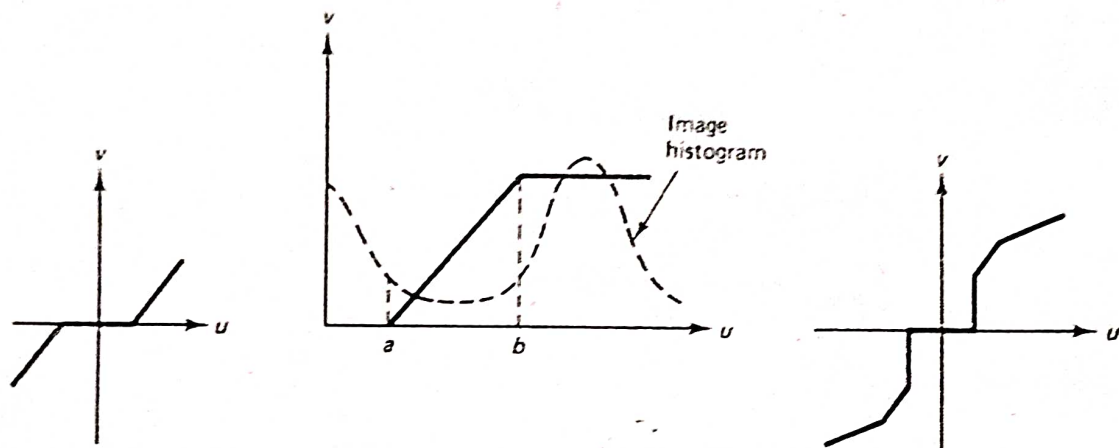


Figure 3: Clipping Transformations.

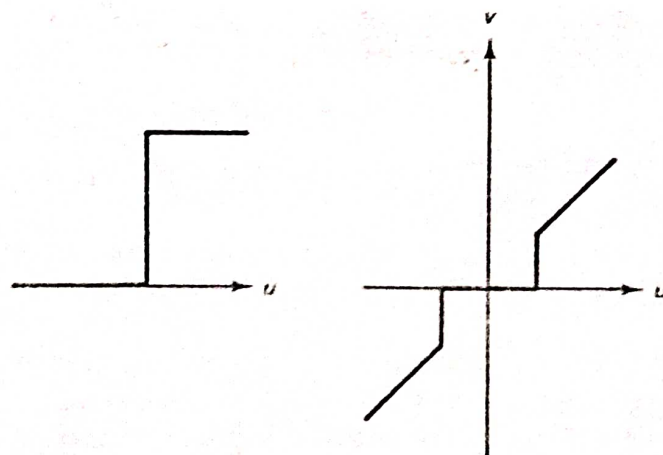


Figure 4: Thresholding Transformations

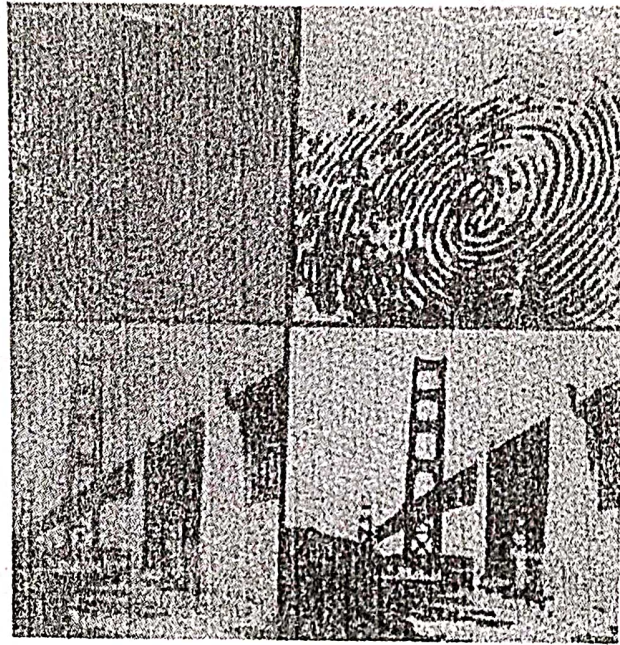


Figure 5 : Clipping and Thresholding Example.

DIGITAL NEGATIVE

A negative image can be obtained by reverse scaling of the gray levels according to the transformation (Figure 6).

$$v = L - u.$$

Figure 7, shows the digital negatives of different images. Digital negatives are useful in the display of medical images and in producing negative prints of images.

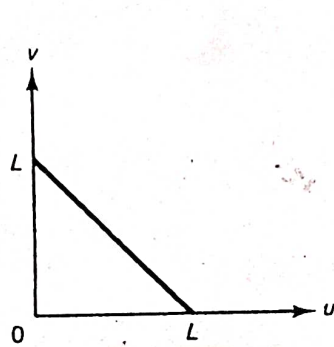
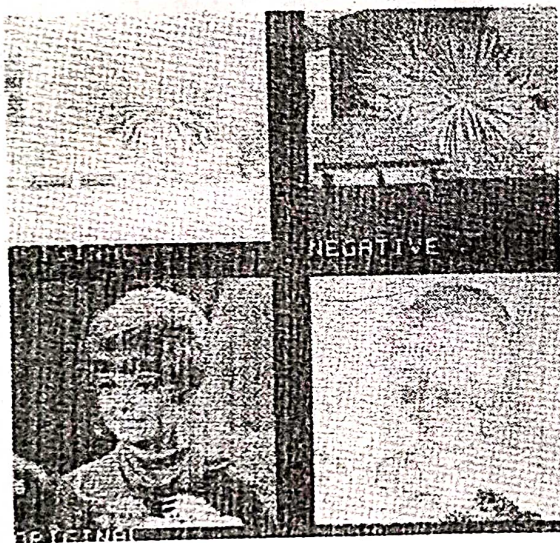
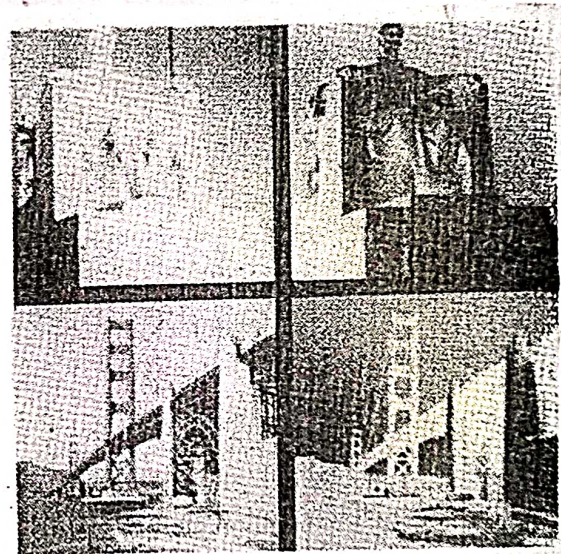


Figure 6 : Digital Negative Transformation



(a)

Figure 7 : Digital negatives.



(b)

Figure 7 : Digital Negatives Example.

INTENSITY LEVEL SLICING

The intensity level slicing can be done either without background or with background.

Without Background:

$$v = \begin{cases} L, & a \leq u \leq b \\ 0, & \text{otherwise} \end{cases}$$

With Background:

$$v = \begin{cases} L, & a \leq u \leq b \\ u, & \text{otherwise} \end{cases}$$

The different intensity level slicings are shown in figure 8.

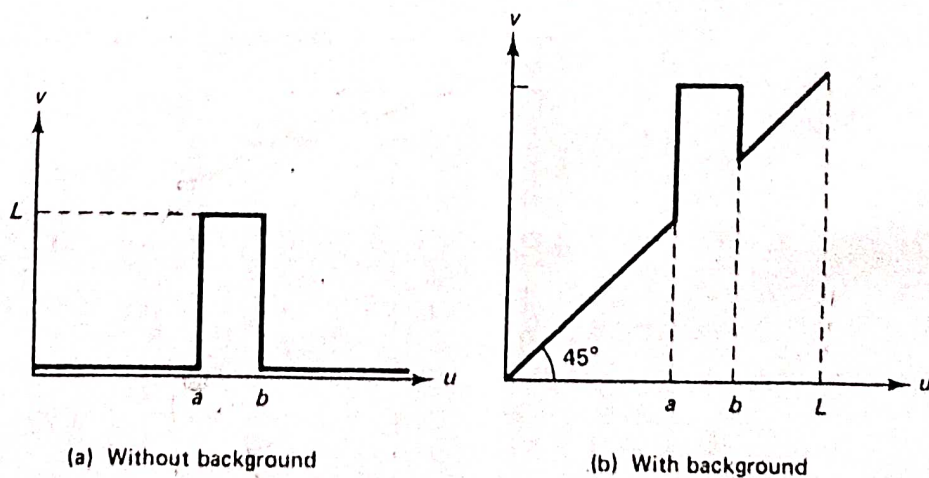


Figure 8: Intensity Level Slicing

These transformations permit segmentation of certain gray level regions from the rest of the image. This technique is useful when different features of an image are contained in different gray levels. Figure 9 shows the result of intensity window slicing for segmentation of low-temperature regions (clouds, hurricane) of two images when high-intensity gray-levels are proportional to low temperatures.

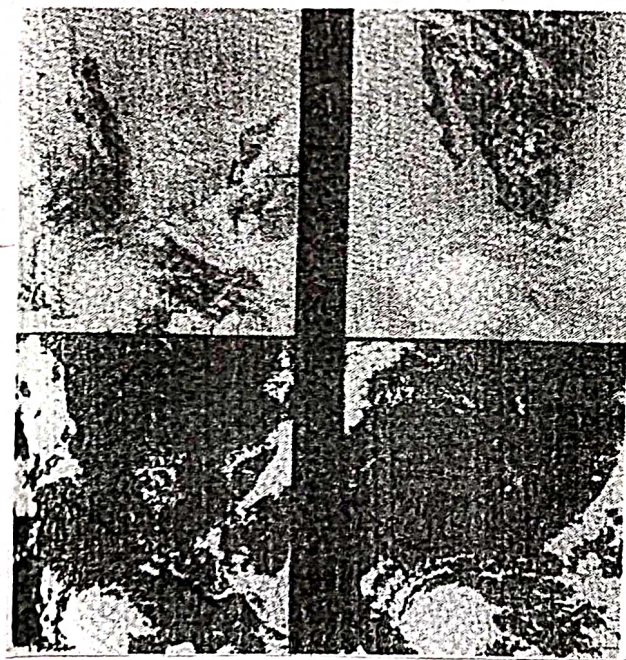


Figure 9: Level Slicing of intensity window $[175, 250]$.
Top row: visual and infrared (IR) images
Bottom row: Segmented images.



BIT EXTRACTION

Suppose each image pixel is uniformly quantized to B bits. It is desired to extract the n^{th} most-significant bit and display it
let

$$u = k_1 2^{B-1} + k_2 2^{B-2} + \dots + k_n 2^{B-n} + \dots + k_{B-1} 2 + k_B$$

Then we want the output to be.

$$v = \begin{cases} 1, & \text{if } k_n = 1 \\ 0, & \text{otherwise} \end{cases}$$

It is easy to show that

$$k_n = i_n - 2i_{n-1}$$

$$\text{Where } i_n \triangleq \text{Int} \left[\frac{u}{2^{B-n}} \right],$$

$$\text{Int}[x] \triangleq \text{integer part of } x.$$

Figure 10 shows the first 8 most-significant bit images of an 8 bit-image. This transformation is useful in determining the number of visually significant bits in an image.

In the figure only the first 8 bits are visually significant, because the remaining bits do not convey any information about the image structure

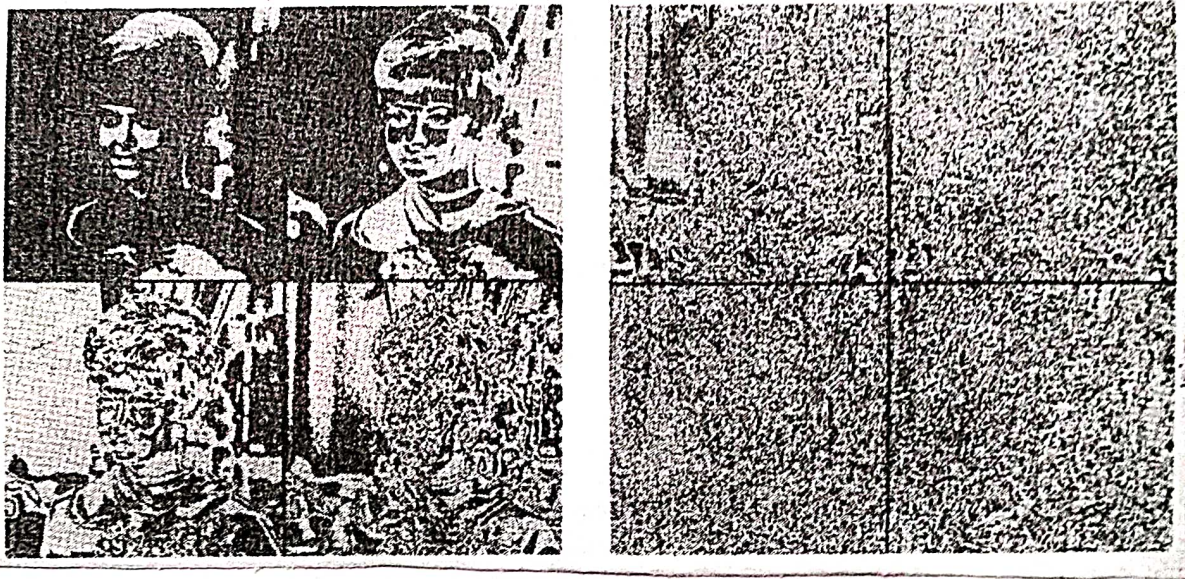


Figure 10: 8 bit planes of a Digital Image

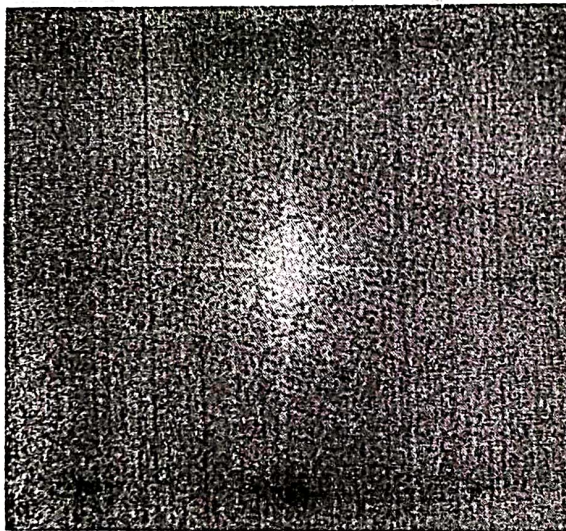
RANGE COMPRESSION

Sometimes the dynamic range of the image data may be very large. For example, the dynamic range of a typically untransformed image is so large that only a few pixels are visible. The dynamic range can be compressed via the logarithmic

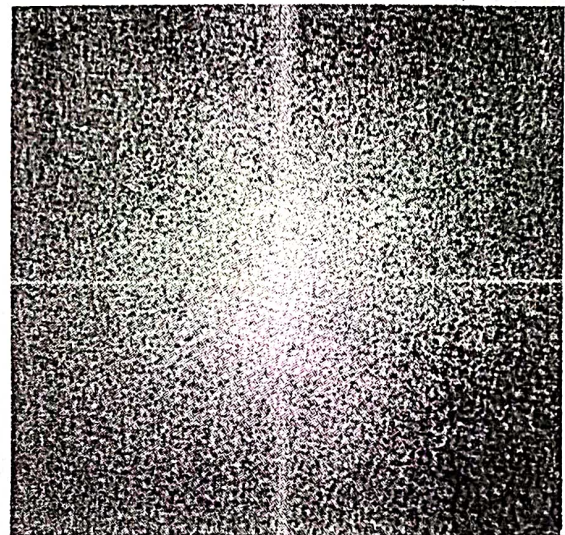
transformation.

$$V = c \log_{10} (1 + |u|)$$

Where c is a scaling constant. This transformation enhances the small magnitude pixels compared to those pixels with large magnitudes. The Figure 11 (a) and (b) shows the original and log of a particular image respectively.



(a) Original



(b) log

Figure 11: Range Compression.

SPATIAL OPERATIONS

Many image enhancement techniques are based on spatial operations performed on local neighborhoods of input pixels. Often, the image is convolved with a finite impulse response filter called spatial mask.

FUNDAMENTALS OF SPATIAL -

CONVOLUTION AND CORRELATION

* Correlation:

Process of moving the filter mask over the image and computing the sum of products at each location.

- Correlation is a function of displacement of the filter.

* First value of correlation corresponds to zero displacement of the filter.

* Second value corresponds to one unit displacement.



- Correlating a filter w with a function that contains all 0s and a single 1 (unit impulse function or discrete unit impulse) yields a copy of w but rotated by 180° .

Convolution:

- * Same as correlation except that the filter is first rotated by 180° .
- * Convoluting a function with a unit impulse yields a copy of the function at the location of the impulse.
 - Pre-rotating the function by 180° and performing the sliding sum of products operations yields the desired result.
 - If the filter mask is symmetric, correlation and convolution yields the same result.

★ Issues of border crossing by the kernel.

- limiting the image to the kernel overlap.
- > limiting the kernel to the image overlap.
- padding image by replication or reflection

Extending the concept to images (2D)

If f contains a region identical to w , value of correlation is maximum when w is centered on that region of f , this property can be used to find the matches between images.

Correlation is given by:

$$w(x,y) \star f(x,y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s,t) f(x+s, y+t)$$

Convolution is given by:

$$w(x,y) \star f(x,y) = \sum_{s=-a}^a \sum_{t=-b}^b w(s,t) f(x-s, y-t)$$

SPATIAL AVERAGING AND SPATIAL LOW-PASS FILTERING

Here each pixel is replaced by a weighted average of its neighborhood pixels, that is

$$v(m,n) = \sum_{(k,l) \in W} a(k,l) y(m-k, n-l) \quad (1)$$

Where $y(m,n)$ and $v(m,n)$ are the input and output images, respectively, W is a suitably chosen window, and $a(k,l)$ are the filter weights. A common class of spatial averaging filters has all equal weights, giving

$$v(m,n) = \frac{1}{N_w} \sum_{(k,l) \in W} y(m-k, n-l) \quad (2)$$

Where $a(k,l) = 1/N_w$ and N_w is the number of pixels in the window W . Another spatial averaging filter used often is given by:

$$v(m,n) = \frac{1}{2} \left[y(m,n) + \frac{1}{4} \{ y(m+1, n) + y(m-1, n) + y(m, n+1) + y(m, n-1) \} \right] \quad (3)$$

Here each pixel is replaced by its average with the average of its nearest four pixels. Figure 12 shows some spatial averaging masks.

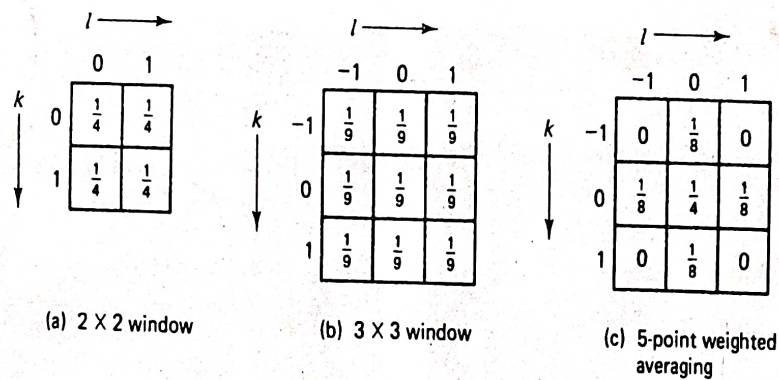


Figure 12: Spatial averaging masks $a(k, l)$

Spatial averaging is used for noise smoothing, low-pass filtering, and sub-sampling of images. Suppose the observed image is given as

$$y(m, n) = u(m, n) + \eta(m, n)$$

(4)

Where $\eta(m,n)$ is white noise with zero mean and variance σ_η^2 . Then the spatial average of Eq (2) yields

$$v(m,n) = \frac{1}{N_w} \sum_{(k,l) \in W} u(m-k, n-l) + \bar{\eta}(m,n) \quad (5)$$

Where $\bar{\eta}(m,n)$ is the spatial average of $\eta(m,n)$

Figure 13 shows examples of spatial averaging of an image containing Gaussian noise.



(a) Original



(b) noisy



(c) 3 × 3 filter



(d) 7 × 7 filter

Figure 13 Spatial averaging filters for smoothing images containing Gaussian noise.

DIRECTIONAL SMOOTHING

To protect the edges from blurring while smoothing, a directional averaging filter can be useful. Spatial averages $v(m, n; \theta)$ are calculated in several directions as shown in Figure 14.

$$v(m, n; \theta) = \frac{1}{N_\theta} \sum_{(k, l) \in W_\theta} y(m-k, n-l) \quad (1)$$

A direction θ^* is found such that $|y(m, n) - v(m, n; \theta^*)|$ is minimum. Then

$V(m, n) = v(m, n; \theta^*)$ gives the desired result. Figure 15 shows an example of this method.

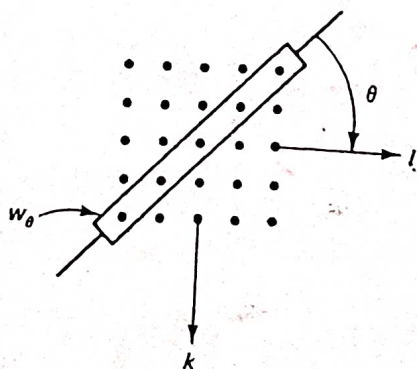
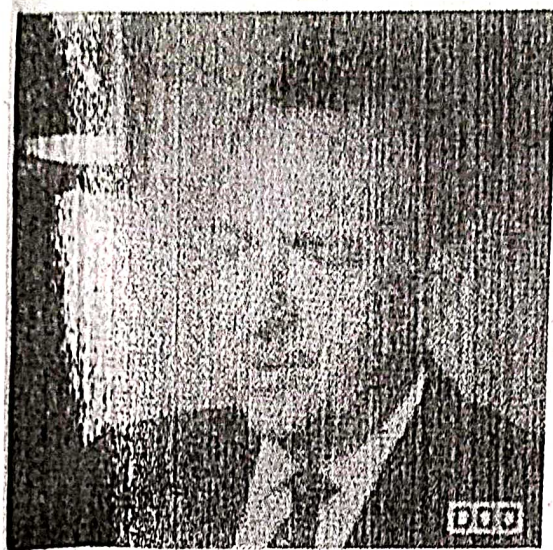
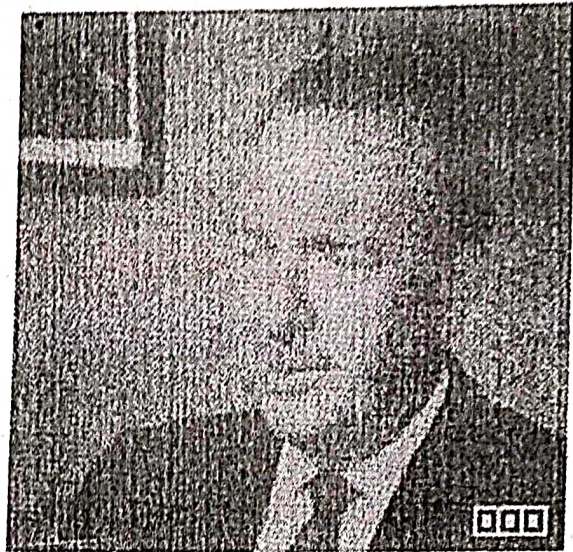


Figure 14: Directional Smoothing Filter.



(a) 5 x 5 spatial smoothing



(b) directional smoothing

Figure 15: Example of Directional Smoothing.

MEDIAN FILTERING

Here the input pixel is replaced by the median of the pixels contained in a window around the pixel, that is,

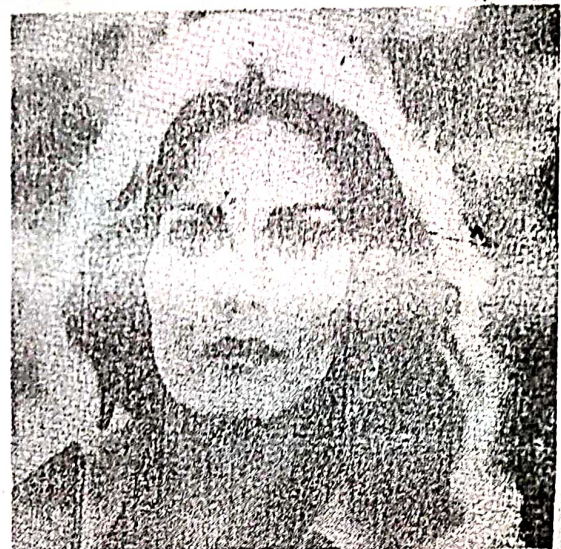
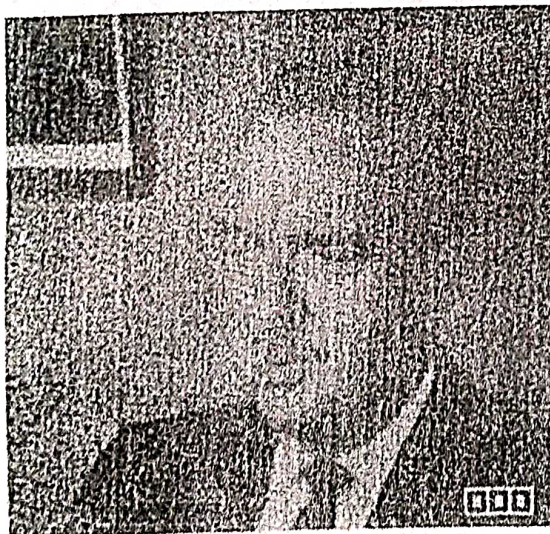
$$v(m, n) = \text{median} \{ y(m-k, n-l); (k, l) \in W \}$$

Where W is a suitably chosen window. The algorithm for median filtering requires arranging the pixel values in the window

in increasing or decreasing order and picking the middle value. Generally the window size is chosen so that N_w is odd. If N_w is even, then the median is taken as the average of the two values in the middle. Typical windows are 3×3 , 5×5 , 7×7 or the five point window considered for spatial averaging in Figure 12C.



(a) Image with binary noise

(b) 3×3 median filtered

(c) Image with Gaussian noise

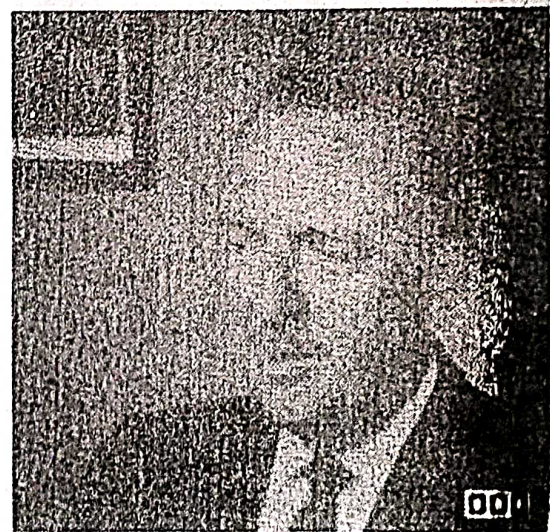
(d) 3×3 median filtered.

Figure 16 : Median Filtering

The median filter has the following properties:

1. It is a non linear filter. Thus for two sequences $x(m)$ and $y(m)$

$$\text{median}\{x(m)+y(m)\} \neq \text{median}\{x(m)\} + \text{median}\{y(m)\}$$

2. It is useful for removing isolated lines or pixels while preserving spatial resolution.

Figure 16 shown above, shows that the median filter performs very well on images containing binary noise but performs poorly when the noise is Gaussian.

3. Its performance is poor when the number of noise pixels in the window is greater than or half the number of pixels in the window.

Since the median is the $(N_w+1)/2$ largest value ($N_w = \text{odd}$), its search requires, $(N_w-1) + (N_w-2) + \dots + (N_w-1)/2 = 3(N_w^2-1)/8$ comparisons. This number equals 30 for 3×3 windows and 224 for 5×5 windows. The no. of comparisons can be reduced approximately $\frac{1}{2} N_w \log_2 N_w$ [5] through

a more efficient search technique.

UNSHARP MASKING AND CRISPENING

The unsharp masking technique is used commonly in the printing industry for crispening the edges. A signal proportional to the unsharp, or low-pass filtered version of the image is subtracted from the image. This is equivalent to adding the gradient, or a high-pass signal, to the image (Shown in figure: 1.7). In general the unsharp masking operation can be represented by

$$v(m,n) = u(m,n) + \lambda g(m,n)$$

Where $\lambda > 0$ and $g(m,n)$ is a suitably defined gradient at (m,n) . A commonly used gradient function is the discrete Laplacian.

$$g(m,n) \triangleq u(m,n) - \frac{1}{4} [u(m-1,n) + u(m,n-1) + u(m+1,n) + u(m,n+1)]$$

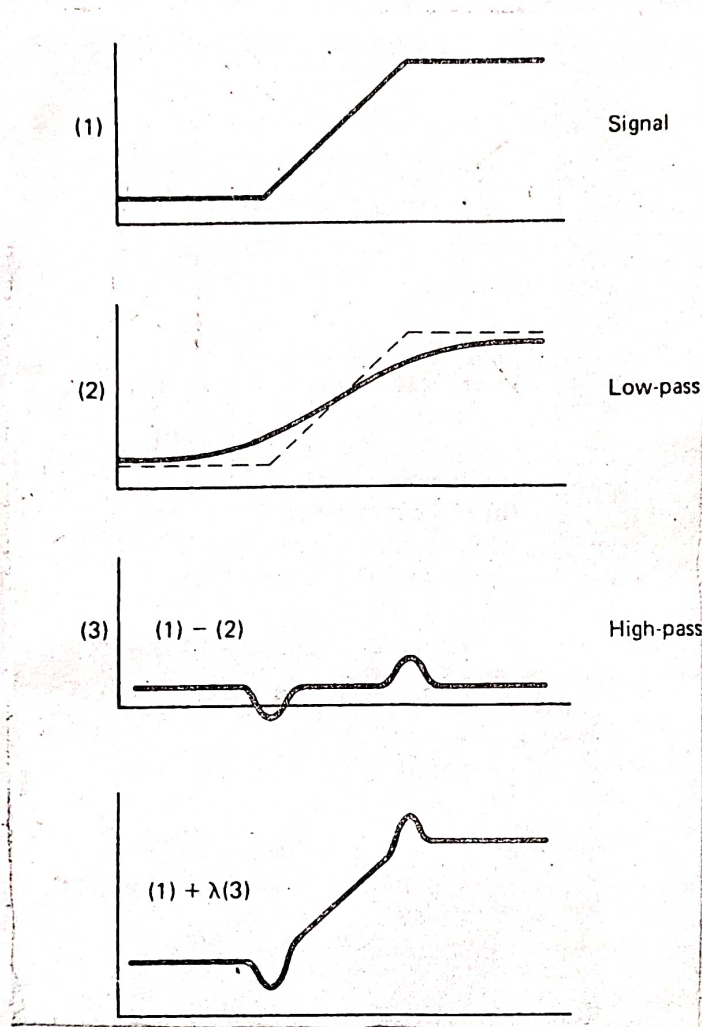


Figure 17: Unsharp masking operations.

BASICS OF FILTERING IN FREQUENCY DOMAIN

Frequency refers to the rate of repetition of some periodic event. In image processing, spatial frequency refers to the variation of image brightness with its position in space. A varying signal can be transformed into a series of simple periodic variations.

If we compute the Fourier Transform of an image and then immediately inverse transform the result, we can regain the same image because a Fourier transform is perfectly reversible. The process of selective frequency inclusion or exclusion is termed frequency domain filtering.

Convolution in spatial domain =

Multiplication in the frequency domain

Filtering in spatial domain = $f(m,n) * h(m,n)$

Where $f(m,n)$ is the input image.

$h(m,n)$ is the filter kernel.

Filtering in the frequency domain = $F(k,l) \times H(k,l)$

Where $F(k,l)$ is the spectrum of the input image.

$H(k,l)$ is the spectrum of the filter kernel.

Thus, frequency domain filtering is accomplished by taking the Fourier transform of the image and the Fourier transform of the image and the Fourier transform of the kernel, multiplying the

the two Fourier transforms, and taking the inverse Fourier transform of the result.

FILTERS IN FREQUENCY DOMAIN

1. LOW-PASS FILTERING IN FREQUENCY DOMAIN

The low-pass filter mask can be broadly classified into two types:

(1) Non-Separable Filter Transfer Function.

The non-separable filter transfer function is given by

$$H(k, l) = \begin{cases} 1, & \text{for } \sqrt{k^2 + l^2} \leq D_0 \\ 0, & \text{otherwise} \end{cases}$$

Where D_0 is the cut-off frequency of the low-pass filter. The cut-off frequency determines the amount of frequency components passed by the filter.

(2) Separable low-Pass Filter Transfer Function

The expression for a separable low-pass filter transfer function is given below

$$H(k, l) = \begin{cases} 1, & \text{for } k \leq D_k \text{ and } l \leq D_l \\ 0, & \text{Otherwise} \end{cases}$$

Butterworth low-Pass Filter

The transfer function of a two-dimensional Butterworth low-pass filter is given by

$$H(k, l) = \frac{1}{1 + \left[\frac{\sqrt{k^2 + l^2}}{D_0} \right]^{2n}}$$

Here, n refers to the filter order and

D_0 represents the cut-off frequency.

2. HIGH-PASS FILTER IN FREQUENCY DOMAIN

The low-pass filters and high-pass filters are complementary to each other. The 2D ideal high-pass filter is one whose transfer function satisfies the relation

$$H(k, l) = \begin{cases} 0, & \text{if } D(k, l) \leq D_0 \\ 1, & \text{if } D(k, l) > D_0 \end{cases}$$

Where D_0 is the cut-off distance measured from the origin of the frequency plane.

The cross section of an ideal high-pass filter is shown in Figure 18. The ideal high-pass filter is not practically realizable.

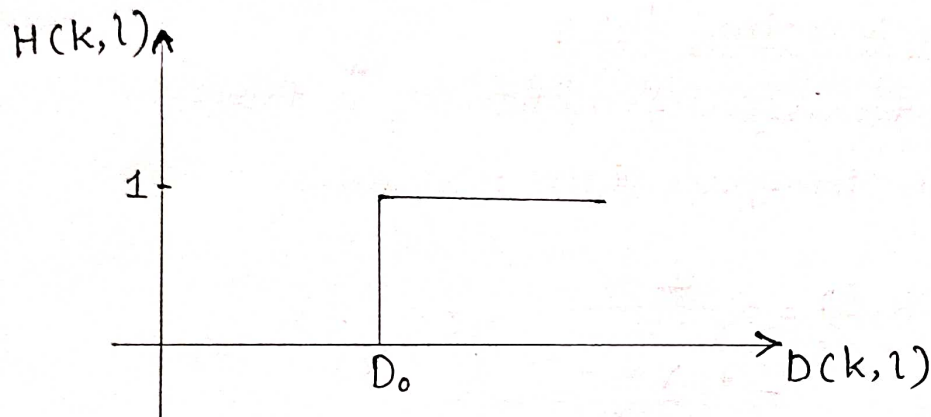


Figure 19: Cross-section of an Ideal high-pass filter.

3. BUTTERWORTH HIGH-PASS FILTER

The transfer function of a two-dimensional Butterworth filter is given by

$$H(k, l) = \frac{1}{1 + \left[\frac{D_0}{\sqrt{k^2 + l^2}} \right]^{2n}}$$

Where 'n' is the order of the filter.

D_0 is the cut-off frequency.

4. LOW-PASS AND HIGH-PASS GAUSSIAN FILTER

The Gaussian filter is popular for the absence of ringing and noise leakage artifacts. The transfer function of a two-dimensional Gaussian low-pass filter is given by.

$$h(m,n) = \frac{1}{2\pi\sigma^2} e^{-\frac{(m^2+n^2)}{2\sigma^2}}$$

The transfer function is separable which can be written as.

$$h(m,n) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{m^2}{2\sigma^2}} \times \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{n^2}{2\sigma^2}}$$

Gaussian High-Pass Filters

The transfer function of the Gaussian high pass filter (GHPF) with cut off frequency locus at a distance D_0 from the center of the frequency rectangle is given by

$$H(u,v) = 1 - e^{-D^2(u,v)/2D_0^2}$$



Where $D(u,v)$ is in the form:

$$D(u,v) = \left[(u - P/2)^2 + (v - Q/2)^2 \right]^{1/2}$$

P and Q are the padded sizes.

SUMMARY OF THE HIGHPASS FILTERS.

Highpass filters, D_0 is the cut off frequency and n is the order of the Butterworth filter.

Ideal High Pass Filter

$$H(u,v) = \begin{cases} 1, & \text{if } D(u,v) \leq D_0 \\ 0, & \text{if } D(u,v) > D_0 \end{cases}$$

Butterworth High Pass Filter.

$$H(u,v) = \frac{1}{1 + [D_0 / D(u,v)]^{2n}}$$

Gaussian High Pass Filter

$$H(u,v) = 1 - e^{-D^2(u,v) / 2D_0^2}$$

Ideal Lowpass Filter

A 2D lowpass filter that passes without attenuation all frequencies within a circle of radius D_0 from the origin and "cuts off" all frequencies outside this circle is called an ideal low pass filter (ILPF).

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

Where D_0 is a positive constant and $D(u,v)$ is the distance between a point (u,v) in the frequency domain and the center of the frequency rectangle; that is

$$D(u,v) = \left[(u-P/2)^2 + (v-Q/2)^2 \right]^{1/2}$$

Where, as before, P and Q are the padded sizes.

The resulting padded images are of size $P \times Q$. If both arrays are of the same size, $M \times N$, then we require that

$$P \geq 2M-1.$$

and

$$Q \geq 2N-1.$$



SUMMARY OF THE LOWPASS FILTERS

In lowpass filters, D_0 is the cutoff frequency and 'n' is the order of the Butterworth filter.

Ideal low Pass Filter

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \leq D_0 \\ 0 & \text{if } D(u,v) > D_0 \end{cases}$$

Butterworth low Pass Filter

$$H(u,v) = \frac{1}{1 + [D(u,v)/D_0]^{2n}}$$

Gaussian low Pass Filter.

$$H(u,v) = e^{-D^2(u,v)/2D_0^2}$$